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Ultrasensitive and highly selective electrochemical determination of mesalazine in pharmaceutical, biological and environmental samples with the use of Co-Mordenite/Mesoporous Carbon modified Glassy Carbon Electrode

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ABSTRACT

In this paper, an ultrasensitive and highly selective analytical method for the voltammetric determination of the anti-inflammatory drug mesalazine (MES) at the Co-exchanged mordenite/mesoporous carbon modified glassy carbon electrode (Co-MOR/MC-GCE) is presented. The combination of unique properties of zeolite and mesoporous carbon, affirmed by conducting morphology and textural parameters studies, resulted in the fabrication of the unique platform sensing, which exhibited a significant enhancement effect of MES oxidation peak current. In the voltammetric determination of mesalazine by the means of differential pulse voltammetry (DPV), univariate optimization stage of the instrumental parameters and the pH value of the supporting electrolyte was performed. In 0.04 mol L^{-1} Britton-Robinson buffer solution (pH 2.0), the oxidation peak of MES at 0.41 V vs. Ag | AgCl | 3 M KCl reference electrode revealed a linear response in the range of 0.99 – 99.01 μg L^{-1} ($6.47{\cdot}10^{-9}$ – $6.47{\cdot}10^{-7}$ mol L^{-1}) with the limit of detection (*LOD*) equal to 0.27 μg L^{-1} ($1.76{\cdot}10^{-9}$ mol L^{-1}). The developed protocol was successfully applied for the direct determination of mesalazine in pharmaceuticals, environmental waters, and biological fluids (human urine and serum) with recovery of 103.4 – 119.3%.

1. Introduction

Mesalazine (MES) or mesalamine, chemically known as 2-hydroxy-5-aminobenzoic acid or 5-aminosalicylic acid, is one of the members of the salicylate drug family used for the treatment of inflammatory bowel disease (IBD). This anti-inflammatory drug is widely administered in the therapy of moderate ulcerative colitis and inflamed rectum, and mild or moderate Crohn's disease [1]. The function of MES is related to a decrease of prostaglandin and leukotriene synthesis, modulating the inflammatory response derived from the cyclooxygenase and lipooxygenase pathways, and protecting cells in the swollen mucosa by scavenging reactive oxygen metabolites and thus reducing their harmfulness [2]. MES can also be used as a gastrointestinal agent to maintain remission and to chemoprevention of colorectal cancer in patients with IBD.

Up to now numerous analytical methods have been proposed for mesalazine determination in tablets and biological fluids. The most commonly applied one is high-performance liquid chromatography (HPLC) combined with, i.e., fluorescence [3], ultraviolet [4], electrochemical [5], and mass spectrometry [6] detection. Other methods that have been successfully applied to the determination of MES are spectrophotometry [7], spectrofluorometry [8], chemiluminescence [9], and capillary electrophoresis [10]. Despite many advantages, the vast majority of these analytical approaches are time-consuming, require the use of high-priced apparatus, involve sophisticated sample preparation, consume large volumes of toxic solvents, and display insufficient detection capability. In view of this, great attention has been paid to the electrochemical determination of mesalazine by means of the voltammetry technique [11], since it offers excellent sensitivity, accuracy, low detection limit, high reproducibility, low cost, and short analysis time. Therefore, in recent years, numerous voltametric sensors of MES with different kinds of the working electrode have been developed [11–13].

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Among these, the most widely used are chemically modified electrodes (CMEs), with a versatile composition of functional nanomaterials introduced to the modification of glassy carbon electrode (GCE) [14–21], disposable screen-printed electrode (SPE) [22,23], and carbon paste electrode (CPE) [24–30].

Nowadays, nanomaterials play a crucial role in the development of electrochemical sensors, improving their stability, sensitivity, and selectivity in the presence of common interference species. Although superior properties, nanomaterials possess some limitations that hinder their application in the electrode's modification. These include physicochemical instability and/or aggregation of nanoparticles, high chemical reactivity, explosive character, biological harmfulness, chemical activation and/or surface functionalization, and difficulties in synthesis related to the application of hazardous and expensive chemicals, and high energy input [31,32]. Hence, the research of new sensing materials has received increasing attention in the field of CMEs.

From the viewpoint of nanomaterial-based sensors, zeolite-modified electrodes (ZMEs) have been of great interest, as shown in a number of literature reviews which have arisen in recent years [33–35]. Zeolites form a wide group of natural or synthetic, micro- and mesoporous, crystalline, hydrated aluminosilicates characterized by unique advantages, valuable in terms of improving electrochemical sensor performance. These particularly include shape, size, and charge selectivity, resulting in molecular sieve properties [36], as well as a high ion-exchange capacity that allows the incorporation of transition metal ions into the tree-dimensional structure and therefore enhanced zeolite electrocatalytic activity [37]. Moreover, zeolites possess high surface area, thermal and chemical stability in acid media, insolubility in most organic solvents, non-toxicity, and simplicity of physical and chemical modification [38]. Zeolite nanostructures with different three-dimensional topology have been widely employed as an electrode modifier for sensitive voltammetric determination of many pharmaceutical compounds [39–45].

In the present study, a highly sensitive and selective voltammetric method of mesalazine determination using a glassy carbon electrode (GCE) modified with a nanocomposite of Co-exchanged form of zeolite with mordenite framework structure (Co-MOR) and mesoporous carbon (MC) as the electron-conducting component is described for the first time. The electrochemical behavior of MES on the Co-mordenite/Mesoporous Carbon modified Glassy Carbon Electrode (Co-MOR/MC-GCE) was investigated by cyclic voltammetry (CV). To provide the highest sensitivity, selectivity, and robustness of MES determination, the optimization of instrumental parameters of differential pulse voltammetry (DPV) and the pH value of the supporting electrolyte was conducted, followed by the interference studies. The analytical parameters, such as linear range, limit of detection, short- and long-term stability, were established and compared with those previously presented in the literature. Finally, the developed protocol was successfully applied for the quantitative analysis of mesalazine in pharmaceuticals, environmental waters, and biological fluids.

2. Experimental

2.1. Chemicals and solutions

Synthetic zeolite powder of Na-mordenite (CVB 10A) was received from Zeolyst International (Pennsylvania, United States). Mesoporous carbon (graphitized carbon) was purchased from Sigma-Aldrich (product number 699624) and used without any pretreatment. A fresh standard stock solution of mesalazine (Pharmaceutical Secondary Standard, Sigma-Aldrich) at a concentration of 1000 mg L^{-1} with the addition of 0.2 mol L^{-1} NaOH was prepared weekly, protected from the light, and stored at 4°C in the refrigerator. Working solutions of MES with lower concentrations were prepared daily by dilution of the standard stock solution with double-distilled water. Britton-Robinson buffer solutions (BRB) with various pH values (2.0 - 4.4) were prepared from an equal (0.04 mol L^{-1}) mixture of CH_3COOH, H_3BO_3, and H_3PO_4 (Avantor Performance Materials Poland S.A) with the appropriate amount of 0.2 mol L^{-1} NaOH (Avantor Performance Materials Poland S.A) solution. Dichloromethane, polystyrene (average M_w~35,000), Triton X-100, sodium dodecyl sulphate (SDS), humic acid (HA), uric acid (UA), cellulose, starch, magnesium stearate, and titanium dioxide were purchased from Sigma-Aldrich. All other chemicals, i.e. acetone, potassium chloride, sodium phosphate, magnesium sulphate, calcium chloride, potassium nitrite, cetrimonium bromide (CTAB), and cobalt(II) nitrate were obtained from Avantor Performance Materials Poland S.A. All reagents were of analytical grade and used as received. The double-distilled water was used to prepare all the aqueous solution. Certified Reference Materials (CRMs) of Surface Water (SPS-SW2 Batch 131) and Waste Water (SPS-WW2 Batch 113) were received from Spectrapure Standards AS, Oslo, Norway. The synthetic serum Nortrol and human urine control (Medidrug Basis-line U) were purchased from Thermo Scientific (Finland) and Medichem Diagnostica GmbH & Co. (Germany), respectively.

2.2. Instrumentation

The voltammetric measurements were realized with the use of the M164 electrode stand combined with the M161 multipurpose electrochemical analyzer (both mtm-anko, Poland). For the real-time data acquisition, the EALab 2.1 software was applied. Throughout the experimental work, a classical three-electrode voltammetric cell with a volume of 10 mL, composed of the bare or modified GCE (MF-2012, φ=3 mm, BASi, USA) working electrode, a self-made double-junction silver-silver chloride (Ag | AgCl | 3 M KCl) reference electrode, and a platinum wire auxiliary electrode, was used. To ensure homogeneity, the solution was stirred between measurements using a magnetic bar coated with Teflon® placed at the bottom of the cell. For the adjustment of pH value of the BRB solution, the SevenCompact S210 laboratory pH-meter (Mettler Toledo, Switzerland) was exploited.

The chemical composition of the initial zeolite material was set out using a wavelength dispersive X-ray fluorescence spectrometer (WD-XRF, PANalytical, Malver, UK) Axios mAX 4 kW equipped with Rh source. For the semiquantitative analysis of the obtained XRF spectra, the Omnian standardless analysis package (PANalyticla) was applied. The textural parameters of mordenite and mesoporous carbon were determined by the multipoint nitrogen adsorption/desorption method at 77 K using the ASAP 2010 instrument (Micrometric Instruments, USA). Samples were degassed at 378 K for 24 h prior to measurement. The specific surface area (SSA) and porosity were calculated with the use of the Brunauer-Emmett-Teller (BET) and the Barret-Joyner-Halenda (BJH) methods, respectively. The Dynamic Light Scattering (DLS) method was implemented with the aim of establishing the particle size distribution (PSD) of MOR and MC. The DLS measurements were conducted with the use of the Zetasizer Nano-ZS (Malvern Instruments, UK) with dedicated software Mastersizer 2000 ver. 5.60. Observation of the morphology of the MOR, MC, and modifying layer was performed using the Tescan Vega3 scanning electron microscope (SEM, ThermoFisher Scientific, USA). Due to employing energy-dispersive X-ray spectroscopy EDS attachment, the elemental mapping and EDS spectra of zeolite before and after the ion exchange with Co^{2+} cations were obtained and used in order to verify the effectiveness of the used modification procedure. Before SEM/EDS measurement, the mordenite sample was sprayed with graphite.

2.3. Zeolite modification and the working electrode fabrication

In the first step, synthetic zeolite – Na-mordenite - was transformed into Co-form via the ion-exchange process according to the procedure previously reported [44]. To replace the Na^+ cations initially presented in the zeolite structure by cobalt(II) ions, 0.5 g of mordenite was treated with 5 mL of 0.1 mol L^{-1} $Co(NO_3)_2{\cdot}6H_2O$ aqueous solution, shaken for

24 h at 25°C, and centrifuged at 13 500 rpm for 10 min. To ensure maximally complete ions replacement, the described procedure was repeated a second time with the fresh cobalt nitrate solution. After that, the zeolite sample was washed three times with a small volume of double-distilled water and dried at room temperature for 24 h. The effectiveness of the used procedure, and thus the receiving Co-form of mordenite (Co-MOR), was confirmed by SEM/EDS measurement.

In the fabrication procedure of the presented sensor, the obtained Co-MOR was mixed with mesoporous carbon (MC) in the weight ratio of 1:1 and ground in an agate mortar. In the next step, 10 mg of the Co-MOR/MC mixture was weighted, transferred to the Eppendorf tube, and connected with 500 μL of the solvent containing 5 mg of polystyrene dissolved in 400 μL of acetone and 600 μL of dichloromethane. The received suspension was homogenized ultrasonically (Ultrasonic bath IS-1, InterSonic, Poland) for 15 minutes. Prior to deposition, the modifying layer, the surface of GCE was polished with 0.3 μm alumina powder (Buehler Micropolish II, USA), washed in water, and rinsed several times in acetone. Finally, 10 μL of homogenized suspension was carefully dropped onto the pretreated surface of GCE and left for complete solvent evaporation at room temperature for 24 h. Thereafter, the fabricated electrode was ready to use. After the measurements, the electrode was carefully rinsed several times in double-distilled water, stored under laboratory conditions (atmospheric air at room temperature) and used in subsequent measurement without any mechanical or electrochemical activation. The very high stability of the modifying suspension and mechanical durability of the modifying layer created the possibility to work with Co-MOR/MC-GCE for many weeks and register up to thousands of voltammetric scans at a given measurement day.

In the way described above, an electrode containing only 10 mg of MC or Co-MOR in the modifying layer was fabricated resulting in mesoporous carbon modified GCE (MC-GCE) and Co-mordenite modified GCE (Co-MOR-GCE), respectively.

2.4. Voltammetric measurements

The electrochemical behavior of mesalazine on the Co-MOR/MC-GCE was established based on the cyclic voltammetry (CV) and differential pulse voltammetry (DPV) measurements in 0.04 mol L^{-1} BRB solution. Cyclic voltammograms were recorded in the presence of 1 mg L^{-1} of MES in the potential range from 0.2 to 0.6 V with the scan rate v changing from 0.0063 to 0.5 V s^{-1}. The DPVs were registered in BRB solution with pH in the range of 2.0 to 4.4 in the presence of 20 μg L^{-1} of MES.

For the calibration of the method, the interference study, and determination of MES in real samples, the DPV technique was implemented. In order to ensure the high selectivity, accuracy, and robustness of the developed protocol, the optimization of instrumental parameters of the DPV was performed. Unless otherwise noted, the DP voltammograms were recorded in 0.04 mol L^{-1} BRB solution (pH 2.0) in the potential range from 0.26 to 0.65 V, under the following optimized parameters: the step potential $E_s = 3$ mV, pulse amplitude $dE = 40$ mV, waiting time $t_w = 30$ ms, and the current sampling time $t_s = 10$ ms. Every measurement was realized in an anodic and subsequently cathodic direction, recording no less than four subsequent scans. The volume of the tested solution was always 5 mL.

2.5. Samples preparation and analysis

The delayed-released dosage tablets - Pentasa® (Ferring Pharmaceuticals GmbH, Germany), as well as gastro-resistant tablets Asamax 250 (QPharma AB, Sweden) containing 500 mg and 250 mg of mesalazine per tablet, respectively, were purchased at a local drugstore. For each drug, five tablets were weighed and ground in an agate mortar into a fine, homogeneous powder. In the amber glass volumetric flask, the MES solution at a concentration of 100 mg L^{-1} was prepared by dissolving an appropriate weight amount of each tablet in 100 mL of double-distilled water and homogenized ultrasonically for 10 minutes. The solutions were diluted 10 times just before measurement. The quantitative determination of MES in each drug was carried out according to the standard addition method. The DP voltammograms were recorded for the blank, after the introduction of diluted samples into a volumetric cell containing 5 mL of 0.04 mol L^{-1} BRB solution with pH 2.0, and for three successive additions of MES standard solution (10 mg L^{-1}). In each stage, four subsequent scans were recorded, and after background correction, the mean value of each peak current was established. The MES content in the analyzed solution was calculated based on the obtained standard addition plot, taking into account the applied dilution factor.

Since the certified reference material of the environmental waters and biological fluids did not contain any mesalazine, the possibility to apply the developed protocol in the analysis of these samples was verified by the recovery test. For this purpose, the CRM of surface and waste water were enriched with an appropriate volume of 1 mg L^{-1} MES standard solution, to obtain a concentration of 20 and 50 μg L^{-1}. After two hours, the spiked samples were subjected to filtration using a syringe filter with a mixed cellulose ester (MCE) membrane and a pore diameter of 0.22 μm (Alchem Grupa Sp. z o.o., Poland). For DPV measurements, 500 μL of the corresponding spiked filtrate was added to 4.5 mL of 0.04 mol L^{-1} BRB solution to obtain the final MES concentration in the voltammetric cell of 2 and 5 μg L^{-1}.

Similarly, synthetic human serum was spiked with the known amount of mesalazine and after two hours was treated with trichloroacetic acid (TCA) to precipitate the high content of proteins present in it [46]. For this purpose, 800 μL of serum was connected with 200 μL of 10% TCA to obtain the final concentration of 2%. Subsequently, the mixture was vortexed for 5 min. and centrifugated for 30 min. at 14 500 rpm. Just before voltammetric measurement, the supernatant was filtered with the use of the MCE membrane syringe filter. 50 μL of the resulting spiked filtrate was transferred to 4.95 mL of supporting electrolyte to obtain the final concentration of MES in the voltammetric cell of 10 and 14 μg L^{-1}.

In the case of synthetic urine, 20 μL of its CRM without any previous preparation was spiked with an appropriate volume of MES standard solution at the concentration of 10 mg L^{-1} and after two hours connected with 4.98 mL of supporting electrolyte (0.04 mol L^{-1} BRB solution) with the aim to obtain the final concentration of MES of 10 and 24 μg L^{-1} in the quartz cell. The DPV measurements of mesalazine in CMRs were carried out according to the standard addition method, including the registration of DP voltammograms for the supporting electrolyte (blank), after adding the appropriate volume of the synthetic spiked samples, and after three standard additions. For each addition, four voltammograms were recorded, averaged, and subjected to background correction. Based on the standard addition plot, the concentration of MES in spiked CRMs was calculated and the recovery parameter was computed.

3. Results and discussion

3.1. Characterization of materials and developed sensors

The results of XRF analysis of the composition of the initial mordenite were as follows (in wt. %): SiO_2 – 79.78, Al_2O_3 – 11.06, Na_2O – 8.48, K_2O – 0.04, MgO – 0.06, CaO – 0.03, TiO_2 – 0.40, Fe_2O_3 – 0.11, and other – 0.09. The established silica to alumina (Si/Al) molar ratio was equal to 12.1. Both mordenite and MC were evaluated with regard to their particle size distribution (PSD) using DLS method. According to the obtained data (Fig. 1A), mordenite exhibits a unimodal PSD with the maximum of 5 μm size fraction, whereas mesoporous carbon is characterized by two maximum size fractions - the first with particle with diameter of about 15 μm of particle size, and the second one with the diameter about 60 μm. Furthermore, the average pore diameter of the modifiers, determined from nitrogen absorption isotherms, confirmed

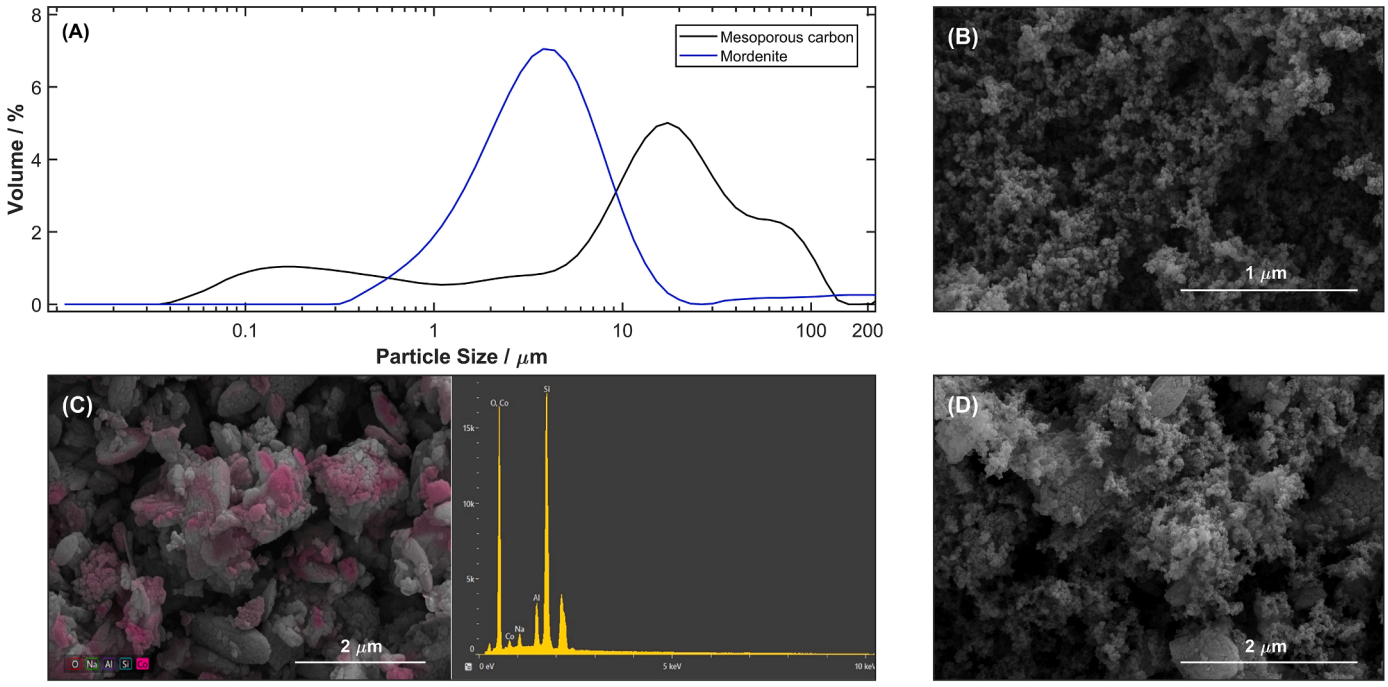


Fig. 1. Material characterization. (A) Particle size distribution of the employed mordenite and mesoporous carbon. (B) SEM images of the mesoporous carbon, (C) Co-mordenite with the corresponding EDS spectrum, and (D) Co-mordenite/MC modifying layer.

the mesoporous nature of the carbon and zeolite material (Table 1). It can be seen that mordenite is characterized by BET surface area and micropore volume (V_{micro}) higher in contrast to MC, which possesses much larger external surface area (S_{ext}) and pore volume.

In the morphology of mesoporous carbon (Fig. 1B) interconnected, irregular, spherical particles can be distinguished, forming aggregates with the size of 1 to 2 μm. SEM observations of mordenite and its Co-form indicate that no crystalline transformations in the zeolite structure occurred during the ion-exchange process with cobalt. As can be seen in Fig. 1C, Co-mordenite reveals the formation of a twinned and intergrown crystal cluster with a flaky habit. Simultaneously, the corresponding EDS spectrum confirms the effectiveness of the incorporation of cobalt ions into the mordenite structure by the presence of the peak connected to Co. The established cobalt content given as the average value of the atomic concentration at several, different surface points amounts to 1.8 at. %. Finally, the morphology of Co-MOR/MC modifying layer is depicted in Fig. 1D. As can be seen, the mordenite crystals seem to be surrounded by aggregates of mesoporous carbon, resulting in a full coating of the GCE surface.

Taking into account the extraordinary physicochemical properties of mordenite and mesoporous carbon, their individual electrochemical performance in the presence of 20 μg L^{-1} of MES in 0.04 mol L^{-1} BRB solution was established (Fig. 2). Before surface modification, the GCE exhibited a relatively high background current with a well-developed oxidation signal of MES at the potential of 0.41 V vs. Ag | AgCl | 3 M KCl reference electrode. In turn, the GCE covered with the layer containing only Co-exchanged mordenite (Co-MOR-GCE) demonstrated practically no electrochemical activity toward mesalazine oxidation. After modification of the GCE with MC (MC-GCE), in the potential range at which the current peak of MES is expected, a small inflection was recorded, resulting in a poor peak obtained after the background subtraction step (inset in Fig. 2). Finally, a significant increase in the analytical signal was achieved when both Co-MOR and MC were introduced into the modifying layer (Co-MOR/MC-GCE). The synergistic effect of the connection of the unique features of zeolite and mesoporous carbon resulted in a more than 14-times higher electrochemical response of Co-MOR/MC-GCE toward mesalazine with respect to bare GCE, thus confirming the legitimacy of its modification.

3.2. Electrochemical behavior of mesalazine on the Co-MOR/CM-GCE

To establish the electrochemical behavior of mesalazine on the Co-MOR/MC-GCE, cyclic voltammetry (CV) measurements at a scan rate from 0.006 to 0.5 V s^{-1} were carried out in the presence of 1 mg L^{-1} of MES in 0.04 mol L^{-1} BRB solution. As can be seen in Fig. 3A, both anodic (solid line) and cathodic (dashed line) peaks of MES oxidation and reduction were registered, respectively. Their presence indicates that the electrode process of mesalazine is electrochemically reversible. Moreover, the potential of anodic (E_{pa}) and cathodic (E_{pc}) peak current remains unchanged with the increase in scan rate, which is an unambiguous sign of the reversible nature of the redox process [47]. Considering the impact of scan rate (ν) on the registered anodic peak current (I_{pa}), the possible diffusion or adsorptive character of the oxidation process of MES was determined. The excellent linear dependence obtained between oxidation peak current and square root of the scan rate value with the regression equation of I_{pa} [μA] = $10.96 \cdot \nu^{1/2}$ [(V

Table 1
Textural properties of zeolite mordenite and carbon mesoporous.

Material	Surface area [m^2 g^{-1}]			Pore volume [cm^3 g^{-1}]		Average pore diameter [nm]
	S_{BET}	S_{micro}	S_{ext}	$V_{micro+mezo}$	V_{micro}	
Mordenite	356.41	320.00	36.41	0.232	0.150	2.61
Mesoporous carbon	200.57	49.70	150.87	0.447	0.022	8.87

S_{BET} - BET surface area; S_{micro} - micropore area; S_{ext} - external surface area; $V_{micro+mezo}$ - micropore+mezopore volume; V_{micro} - micropore volume.

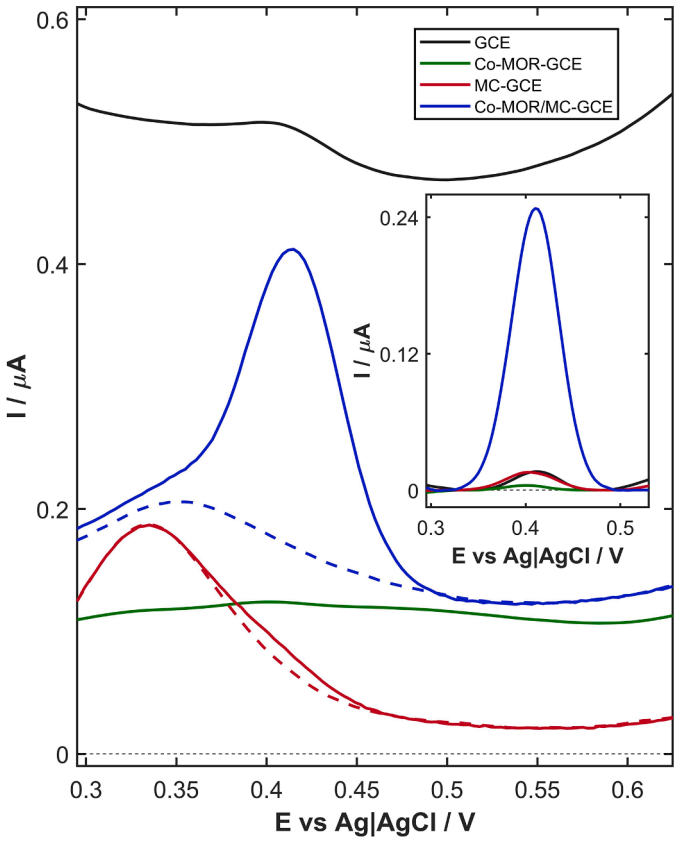


Fig. 2. Comparison of DP voltammograms recorded for 20 μg L^{-1} MES ($1.3{\cdot}10^{-7}$ mol L^{-1}) on the bare GCE, Co-mordenite (Co-MOR-GCE), mesoporous carbon (MC-GCE), and Co-mordenite/mesoporous carbon modified GCE (Co-MOR/MC-GCE). Dashed lines represent the background signal. Inset: DP voltammograms subjected to the background correction. Experimental conditions: E_p = 0.26 V, E_k = 0.65 V, E_s = 3 mV, dE = 40 mV, t_w = 30 ms, t_s = 10 ms. Supporting electrolyte: 0.04 mol L^{-1} BRB solution (pH 2.0).

$s^{-1})^{1/2}$] - 0.826 (R = 0.995) (not showed), indicates that oxidation of MES on the Co-MOR/MC-GCE surface is a diffusion-controlled process. The number of electrons n taking part in the electrochemical oxidation of mesalazine can be established based on the difference between the anodic peak potential E_{pa} and the potential $E_{pa/2}$ at which the current is at half the peak value according to the equation [47]:

$$|E_{pa} - E_{pa/2}| = 56.5 / n \text{ mV} \quad (1)$$

The calculated $|E_{pa} - E_{pa/2}|$ for the analyzed reaction was equal to 31 mV, which gave the electron number value equal to 1.82, indicating the participation of two electrons in the MES oxidation process.

Furthermore, to obtain complete information about the mesalazine oxidation process, the influence of the pH of the supporting electrolyte on the E_{pa} was examined. For each pH value ranging from 2.0 to 4.4 of BRB solution (0.04 mol L^{-1}), four consecutive scans were recorded in the absence (background signal) and in the presence of 20 μg L^{-1} of MES. As a result of the baseline correction step, which involved the subtraction of the registered background signal from a voltammogram recorded in the presence of MES in the voltammetric cell, the curves presented in the Figure 3B were obtained.

As shown, the peak potential shifted toward a more negative value with the increasing pH value of the supporting electrolyte, which indicates the involvement of protons in the MES oxidation reaction. The dependence between E_{pa} and pH was linear (Fig. 3C) with a slope of (-0.058 ±0.002) V per unit change in pH, which is close to the theoretical value of 0.059 V pH^{-1} characteristic for the electrode process with the same number of protons and electrons exchanged during the reaction. Thus, on the basis of the conducted research, it can be concluded that two electrons and two protons participate in the mesalazine electrode reaction. The reversible electrochemical oxidation of MES on the Co-MOR/MC-GCE to quinone-imine form, according to the mechanism depicted in Fig. 3C, is consistent with previous literature reports [13].

According to the DP voltammograms presented in Fig. 3B, the pH of the supporting electrolyte influences not only the MES oxidation peak potential, but also the peak current. It is clearly visible that the current response of Co-MO/MC-GCE increased rapidly with the decrease of the pH value, reaching its maximum at pH 2.0. Thus, to provide the most favorable performance of the sensor for MES determination, the BRB solution with this pH value was selected for further measurements.

3.3. Optimization of the experimental parameters

The influence of the measurement condition on the recorded oxidation signal of mesalazine was established based on the univariate optimization procedure performed for 20 μg L^{-1} MES in 0.04 mol L^{-1} BRB solution (pH 2.0). In the optimization study, the instrumental parameters of the DPV technique, such as step potential (E_s) from 1 to 6 mV, pulse amplitude (dE) in the range of 10 to 60 mV (both positive and negative values), current sampling time (t_s) from 5 to 40 ms, and waiting time (t_w) in the range of 5-40 ms, were systematically varied. The optimal conditions for MES determination using DPV, selected by the comparison of shape, position, symmetry, and height of the peak current, as well as the signal-to-noise ratio, were found as follows: E_s = 3 mV, dE = 40 mV, t_s = 10 ms, and t_w = 30 ms.

3.4. Analytical characteristics of developed method

To investigate the analytical performance of the Co-MOR/MC-GCE, the DP voltammograms were registered for different concentrations of MES in 0.04 mol L^{-1} BRB solution under the optimized conditions. The linear response of the proposed sensor was established based on three independent calibration studies with the use of three different copies of the fabricated electrode and three different additions of the standard solution of MES at the concentration of 1, 5, and 10 mg L^{-1}. From every voltammogram, the background signal recorded in the supporting electrolyte in the absence of the depolarizer (dashed line in Fig. 4A) was subtracted. Based on the obtained curves the calibration graphs were plotted and presented in Figure 4. The coefficients of the calibration curves and the validation parameters computed on their basis, such as linear range (*LR*), limit of detection (*LOD*), and limit of quantification (*LOQ*), were summarized in Table 2.

As shown, the Co-MOR/MC-GCE demonstrated an excellent linear correlation between the oxidation peak current and the MES concentration in all tested different concentration ranges - the first from 0.99 to 11.86 μg L^{-1} (Fig. 4C), the second from 4.99 to 39.68 μg L^{-1} (Fig. 4A), and the third from 9.99 to 99.01 μg L^{-1} (Fig. 4B) with correlation coefficient exceeding 0.9990. It should be underlined that the slopes obtained for each calibration curve are comparable, which indicated that Co-MOR/MC-GCE exhibited a wide linear response covering all testes concentration ranges. Moreover, nearly the same sensitivity achieved for three separate copies of the sensor clearly reveals its high reproducibility in terms of the fabrication procedure. The values of *LOD* and the *LOQ*, calculated according to the following equation: $LOD = 3SD/b$, and $LOQ = 10SD/b$, where *SD* is the standard deviation of y-intercepts of the regression line and b is its slope, were estimated to 0.27 μg L^{-1} ($1.76{\cdot}10^{-9}$ mol L^{-1}) and 0.9 μg L^{-1} ($5.88{\cdot}10^{-9}$ mol L^{-1}), respectively.

The obtained analytical parameters were compared with those reported in the literature for other voltammetric methods for the determination of mesalazine (Table 3). The presented data clearly show that the Co-MOR/MC-GCE is characterized by the lowest limit of detection value among electrochemical sensors described so far. In contrast to other chemically modified electrodes, the simple drop-casting method of

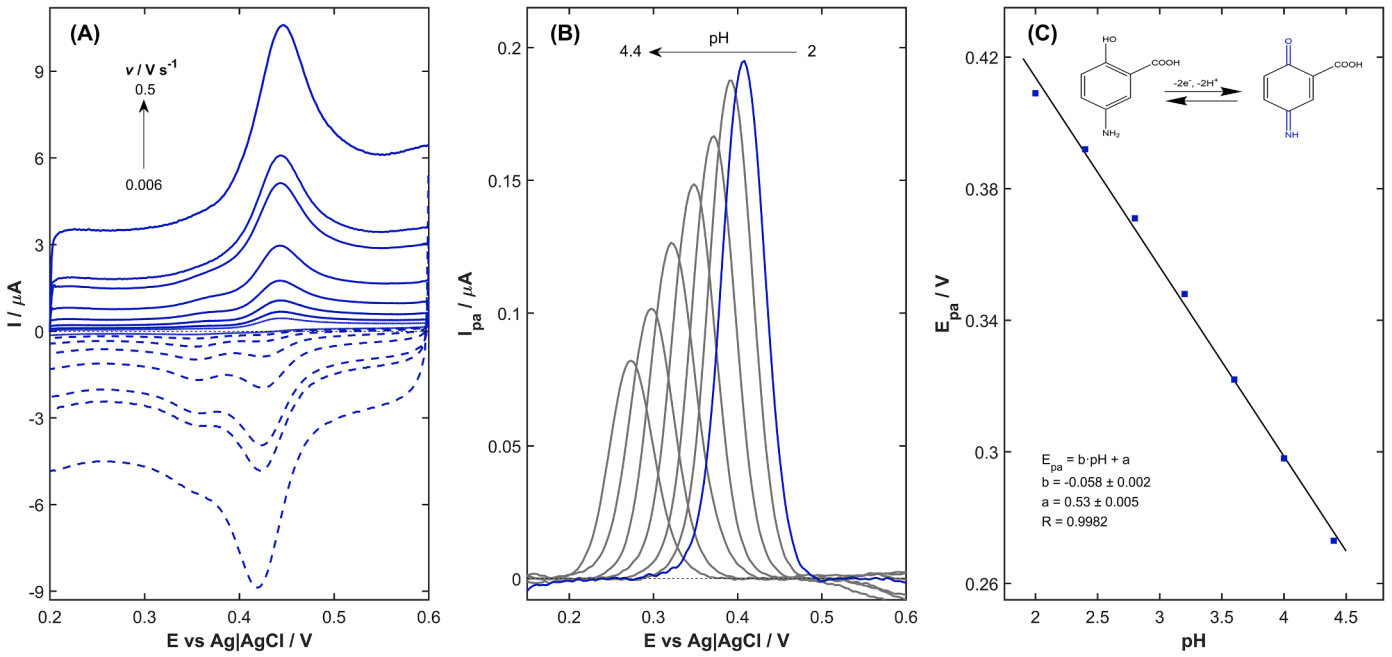


Fig. 3. (A) Cyclic voltammograms recorded for 1 mg L^{-1} MES in 0.04 mol L^{-1} BRB solution (pH 2.0) on Co-MOR/MC-GCE using increasing scan rate values v (given in the legend). Solid lines represent the anodic scan, whereas the dashed ones – the cathodic scan. (B) DPVs registered for oxidation of 20 μg L^{-1} MES in 0.04 mol L^{-1} BRB solution of pH ranging from 2.0 to 4.4 after background subtraction. (C) The dependence between the peak potential (E_{pa}) and pH of the supporting electrolyte. Inset: the mechanism of mesalazine oxidation.

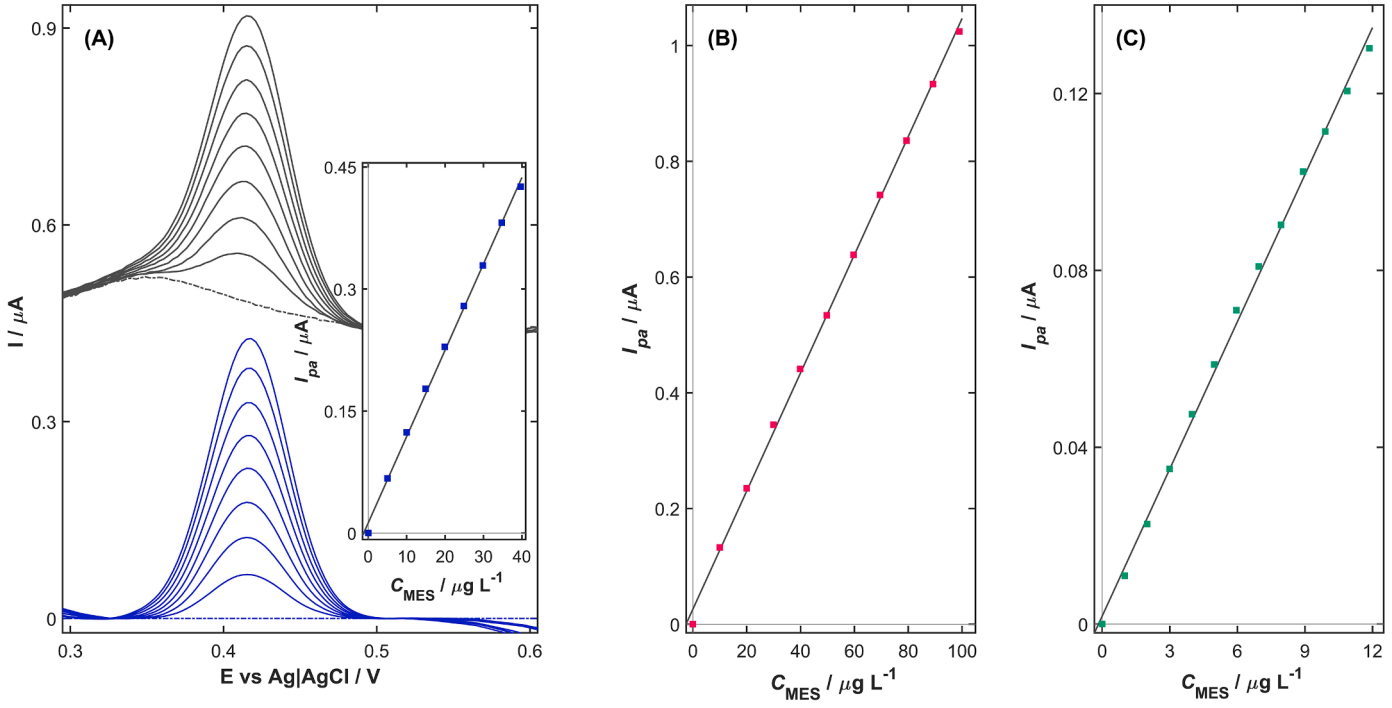


Fig. 4. (A) DP voltammograms registered on the Co-MOR/MC-GCE for the increasing concentration of MES in the range from blank to 40 μg L^{-1} (black lines) with the curves obtained after the background subtraction (blue lines). Inset: corresponding calibration plots. Calibration plot with the regression equation for the MES concentration raging from blank to (B) 99.01 μg L^{-1} and (C) 11.86 μg L^{-1}. Experimental conditions as in Fig. 2.

the fabrication of the proposed sensor does not require any pretreatment processes of mesoporous carbon, as well as utilization of harmful solvents, and expertise instrumentation. Moreover, since the durability and long-term stability of modifying layer, the Co-MOR/MC-GCE can be used in considerable numbers of measurements. The exceptional repeatability and short-term stability of the proposed sensor can be confirmed by the RSD value of peak current recorded for 20 μg L^{-1} MES (three consecutive scans) on ten separate copies of the electrode, which did not exceed 2%. On the other hand, the RSD of the peak currents obtained at all tested copies equals 6% demonstrates the very good reproducibility of the Co-MOR/MC-GCE modification procedure.

Table 2
Analytical parameters obtained for the developed procedures for mesalazine determination.

Parameter	Unit	Fig. 4A	Fig. 4B	Fig. 4C
Tested concentration range	μg L^{-1} (mol L^{-1})	4.99–39.68 ($3.26{\cdot}10^{-8}$–$2.59{\cdot}10^{-7}$)	9.99–99.01 ($6.52{\cdot}10^{-8}$–$6.47{\cdot}10^{-7}$)	0.99–11.86 ($6.47{\cdot}10^{-9}$–$7.74{\cdot}10^{-8}$)
Intercept *a*	μA	0.013 ± 0.004	0.026 ± 0.006	0.002 ± 0.001
Slope *b*	μA L μg^{-1}	0.0106 ± 0.0002	0.0102 ± 0.0001	0.0111 ± 0.0001
R	–	0.9990	0.9995	0.9990
LOD	μg L^{-1} (mol L^{-1})	1.13 ($7.39{\cdot}10^{-9}$)	1.76 ($1.15{\cdot}10^{-8}$)	0.27 ($1.76{\cdot}10^{-9}$)
LOQ	μg L^{-1} (mol L^{-1})	3.77 ($2.46{\cdot}10^{-8}$)	5.88 ($3.84{\cdot}10^{-8}$)	0.90 ($5.88{\cdot}10^{-9}$)

Table 3
Comparison of the voltammetric methods developed for mesalazine determination in different samples.

Electrode	Technique	Linear range / mol L^{-1}	LOD / mol L^{-1}	Samples	Ref.
[1]GCE	LSV	$1{\cdot}10^{-4}$ – $2{\cdot}10^{-6}$	$8.16{\cdot}10^{-7}$	Pharmaceuticals	[11]
[2]PGE	SWV	$9.78{\cdot}10^{-7}$–$7.25{\cdot}10^{-5}$	$2.12{\cdot}10^{-8}$	Tablets	[12]
[3]BDDE	SWV	$2{\cdot}10^{-6}$ – $3{\cdot}10^{-4}$	$7{\cdot}10^{-7}$	Tablets, human urine	[13]
[4]1,5-NDSA doped-PPY/CNT/ GCE	LSV	$1{\cdot}10^{-8}$ – $1{\cdot}10^{-6}$	$3{\cdot}10^{-9}$	Tablets, human plasma	[14]
[5]CNTs-N/GCE	SWV	$5{\cdot}10^{-7}$–$2.5{\cdot}10^{-5}$	$1.1{\cdot}10^{-7}$	Tablets, human serum	[15]
[6]GO/MIP/GCE	DPV	$2{\cdot}10^{-6}$–$2{\cdot}10^{-5}$ $2{\cdot}10^{-5}$–$1.5{\cdot}10^{-4}$	$9.7{\cdot}10^{-7}$	Tablets	[16]
[7]MGCE	CV	$50{\cdot}10^{-6}$–$0.5{\cdot}10^{-3}$	$2.394{\cdot}10^{-8}$	–	[17]
[8]CNT-NH_2/chitosan/GCE	SWV	$1{\cdot}10^{-6}$–$1.8{\cdot}10^{-5}$	$2.1{\cdot}10^{-7}$	Tablets, serum	[18]
[9]PMB_{DES}/CNT/ GCE	DPV	$5{\cdot}10^{-6}$–$1{\cdot}10^{-4}$	$5.8{\cdot}10^{-8}$	Pharmaceutical	[19]
[10]CNT/PMBDES/GCE	DPV	$5{\cdot}10^{-6}$–$1{\cdot}10^{-4}$	$7{\cdot}10^{-9}$	Pharmaceutical	[19]
[11]PMAMGCE	CV	$2.04{\cdot}10^{-6}$–$8.905{\cdot}10^{-6}$	$7.95{\cdot}10^{-8}$	–	[20]
[12]Poly(MX)MGCE	CV	$2.5{\cdot}10^{-4}$–$12.5{\cdot}10^{-4}$	$9.1{\cdot}10^{-5}$	Tablets	[21]
[13]g-C_3N_4/$CeVO_4$/SPCE	DPV	$2{\cdot}10^{-9}$–$3.8{\cdot}10^{-4}$	$5.47{\cdot}10^{-9}$	Tablet, human serum, urine, river water	[22]
[14]ZnCr-LDH/WC	DPV	$3{\cdot}10^{-8}$–$2.53{\cdot}10^{-4}$	$6{\cdot}10^{-9}$	Human urine, river water	[23]
[15]CTAB/MCPE	CV	$6{\cdot}10^{-5}$–$1.4{\cdot}10^{-4}$	$1.9{\cdot}10^{-9}$	Pharmaceutical	[24]
[16]PBQC/CPE	DPV	$2{\cdot}10^{-6}$–$6{\cdot}10^{-4}$	$7{\cdot}10^{-8}$	Pharmaceutical	[25]
[17]PA/CPE	CV	$20{\cdot}10^{-6}$–$0.18{\cdot}10^{-3}$	$2.3{\cdot}10^{-8}$	Tablet	[26]
[18]SDS/CPE	DPV	$1{\cdot}10^{-6}$–$7{\cdot}10^{-7}$	$2.38{\cdot}10^{-7}$	Tablet	[27]
[19]ZIF-67/CPE	DPV	$3{\cdot}10^{-8}$–$5{\cdot}10^{-5}$	$1{\cdot}10^{-8}$	Human urine, serum	[28]
[20]PCR-CPE	CV	$8{\cdot}10^{-5}$–$2{\cdot}10^{-4}$	$1.12{\cdot}10^{-7}$	Tablets	[29]
[21]Co/Cu-BTC@SiO_2/CPE	DPV	$5{\cdot}10^{-8}$–$2{\cdot}10^{-4}$	$1{\cdot}10^{-7}$	Human plasma	[30]
Co-MOR/MC-GCE	DPV	$6.47{\cdot}10^{-9}$–$6.47{\cdot}10^{-7}$	$1.76{\cdot}10^{-9}$	Tablets, human serum, urine, surface and river water	This work

[1] Glassy Carbon Electrode,
[2] Penicil Graphite Electrode,
[3] Boron-doped diamond electrode,
[4] Multiwalled Carbon Nanotubes Poly-pyrrole doped by 1,-Naphtalenedisulfonic Acid Modified GCE,
[5] Multi-walled carbon nanotubes modified GCE,
[6] Graphene oxide coated with an imprinted silica sol–gel modified GCE,
[7] Poly(glutamic acid) modified GCE,
[8] Carbon nanotubes functionalized with amino groups and chitosan modified GCE,
[9] Poly(Methylene Blue in Ethaline Deep Eutectic Solvent/Carbon nanotubes modified GCE,
[10] Multiwalled Carbon Nanotubes/Poly(Methylene Blue in Ethaline Deep Eutectic Solvent/ modified GCE,
[11] Phosphomolybdic acid modified GCE,
[12] Poly-murexide modified GCE,
[13] Graphitic-carbon nitride/cerium vanadate screen printed carbon paste electrode,
[14] ZnCr layered double hydroxide (LDH)/tungsten carbide screen-printed carbon electrode,
[15] Cationic surfactant-CTAB modified carbon paste electrode,
[16] Poly (benzoquinone) chromium(III) complex carbon paste electrode,
[17] Poly(alanine) carbon paste electrode,
[18] Sodium dodecyl sulfate modified carbon paste electrode,
[19] Zeolitic imidazole frameworks- 67 modified carbon paste electrode,
[20] Polymerized-congo red modified carbon paste electrode,
[21] Bimetallic Co/Cu- benzene-1,3,5-tricarboxylic acid@SiO2 modified carbon paste electrode. LSV - linear sweep voltammetry, SWV - square wave voltammetry, DPV - differential pulse voltammetry, CV - cyclic voltammetry.

3.5. Interference studies

To verify the selectivity of the Co-MOR/MC-GCE, the influence of numerous potential interferents, which are usually present in the matrix of pharmaceutical, biological, and environmental samples, on the MES oxidation signal was investigated. Based on them, for the interference studies, inorganic ions (K^+, Na^+, Mg^{2+}, Ca^{2+}, Cl^-, NO_3^-, PO_4^{3-}, and SO_4^{2-}) in the concentration of 0.1 mol L^{-1} were systematically added to supporting electrolyte containing 20 μg L^{-1} of MES, resulting in 10 to 2000-fold excess of particular ions in comparison to the concentration of mesalazine. Similarly, the possible interference effect of surface-active compounds, i.e. sodium dodecyl sulphate (SDS), cetrimonium bromide (CTAB), and Triton-X100, as well as humic acid (HA), uric acid (UA) at the concentration ranging from 0.5 to 20 mg L^{-1}, was tested. To establish the influence of titanium dioxide, starch, magnesium stearate, and cellulose on the recorded signal of 20 μg L^{-1} MES, from 1 to 5 mg of powdered substances was separately added directly to the voltammetric cell. In each case, it was assumed that a given specie has little or no effect on the analytical signal when the percentage difference between the MES peak current registered in the absence and presence of the individual interferent is less than ± 5%.

The obtained results (not shown) clearly indicate that all tested

inorganic ions do not exhibit any interference effect on the mesalazine signal – upon their addition, no significant changes in background current, as well as MES peak shape and position, were observed. The same findings were made with regard to the addition of the basic pharmaceutical excipients, i.e., titanium dioxide, starch, magnesium stearate, and cellulose, the presence of which caused a small divergence in the heigh of the analytical signal, however not exceeding the ±5% tolerance limit. A more noticeable interference effect was noted in the case of surface-active compounds, humic and uric acid. With the successive addition of Triton-X100 (non-ionic surfactant) and CTAB (cationic surfactant), as well as UA, the recorded peak current decreased, reaching respectively 93%, 82%, and 67% of its initial value in the case of the highest concentration of these interferents (20 mg L^{-1}). In turn, the presence of SDS (anionic surfactant) and humic acid at the concentration of 20 mg L^{-1} leads to an increase of the MES oxidation signal by 27% and 19%, respectively. It is important to emphasize that the negative impact of the above-mentioned interferents on the peak height is observed for 1000-fold excess in comparison to trace concentration of MES (20 μg L^{-1}), while simultaneously maintaining the shape and position of the signal. Thus, it can be concluded that the developed method is characterized by the high selectivity required during the determination of real samples with complex matrices.

3.6. Analysis of the real samples

Taking into account the possible side effect associated with long-term treatments of MES, as well as the rigorous requirements of drug quality control, the developed procedure was applied for the determination of mesalazine in real samples of the significance in both the clinical, pharmaceutical and environmental aspects (Fig. 5). Their analysis were conduced according to the standard addition method (Section 2.5) with the subsequent baseline correction step, which was performed by subtracting the background signal recorded in the supporting electrolyte in the absence of the depolarizer (dashed line in Figure 5) from every curves.

In the case of medications (Fig. 5A) and CMR of surface water spiked with MES (Fig. 5B), no other peaks apart from the analytical signal of MES were observed, which indicate high selectivity of the proposed protocol for MES determination. Regarding CMRs of human serum (Fig 5C) and urine, apart from the signal related to the oxidation of MES at +0.41 V, an additional peak at +0.49 V, corresponding to the presence of uric acid (UA) in the tested fluids was observed. Despite its occurrence, the increase of MES peak current after each standard addition was observed, proving that the influence of matric ingredients does not prevent the direct determination of mesalazine in biological fluid with the use of the proposed strategy.

The computed MES contents in the tested pharmaceuticals and recovery parameters obtained for CRMs of biological fluids and environmental waters are summarized in Table 4.The results obtained from the analysis of content analysis in tested medications are consistent with the manufacturer's declaration, as proven by the relative error (RE) to the extent of 6% and demonstrate very good precision confirmed by the RSD value below 2%. In relation to samples spiked with MES at two different concentrations, the resulting recoveries ranged from 103.4% to 119.3% and RSD does not exceed 5%, justifying the suitability of the Co-MOR/

Table 4
Results of mesalazine determination in the pharmaceuticals and CRMs of biological fluids and environmental waters.

Sample	Amount of MES [mg per tablet]		RE [%]	RSD [%]
	Declared	Found $\bar{x} \pm s$		
Pentasa	500	531 ± 24	6.1	4.5
Asamax	250	260 ± 12	3.9	4.6
Sample	**Concentration of MES [μg L^{-1}]**		**Recovery [%]**	**RSD [%]**
	Added	Found $\bar{x} \pm s$		
Surface Water	2	2.09 ± 0.06	104.5	2.8
SPS-SW2	5	5.17 ± 0.18	103.4	3.5
Waste Water	2	2.09 ± 0.05	104.5	2.4
SPS-WW2	5	5.39 ± 0.11	107.8	2.0
Serum	10	11.3 ± 0.21	113.0	1.9
	14	16.7 ± 0.02	119.3	0.1
Urine	24	26.2 ± 0.77	109.2	3.0
	10	11.6 ± 0.45	116.0	3.9

$\bar{x}$ – mean value; s – standard deviation; RSD% = $s/\bar{x} \cdot 100\%$

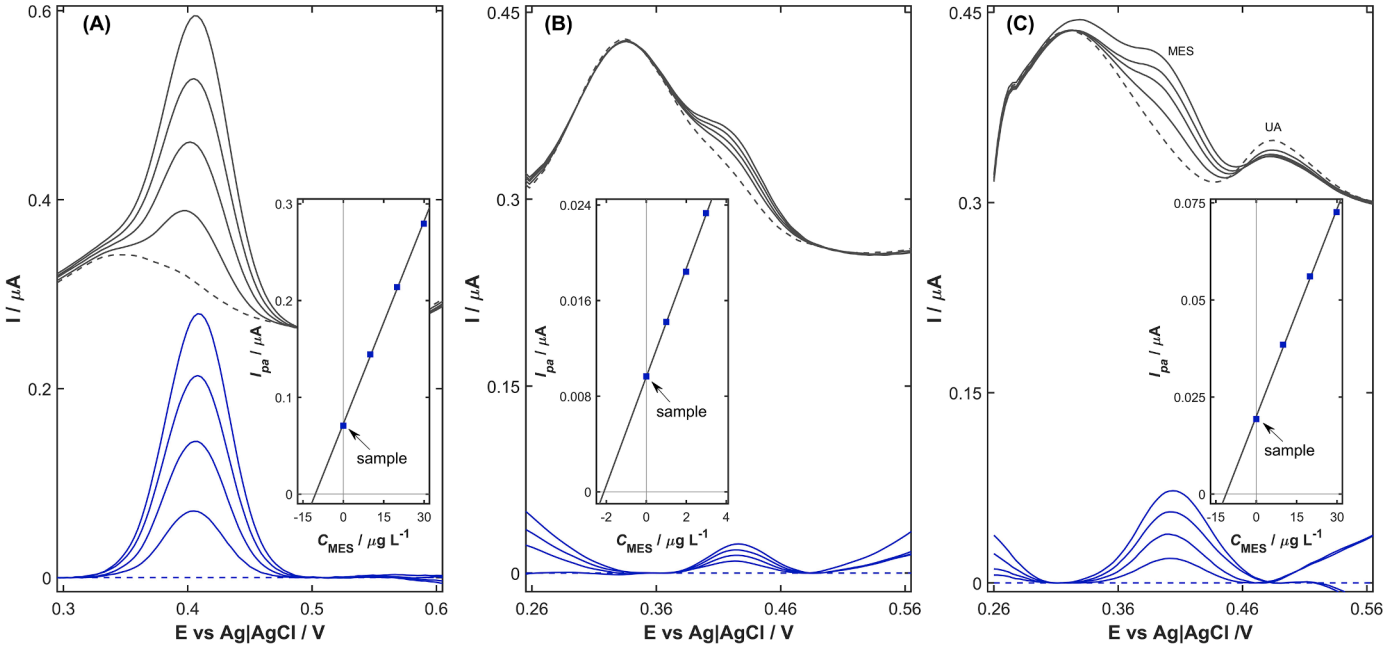


Fig. 5. Investigation of the real samples: (A) gastro-resistant tablets Asamax 250, (B) wastewater CRM spiked with 2 μg L^{-1} MES, and (C) CMR of human serum spiked with 10 μg L^{-1} MES. The experimental voltammograms were depicted in black, whereas the curves obtained after the background correction are marked in blue. Inset: corresponding calibration plots. Experimental conditions as in Fig. 2.

MC-GCE for the quantitative determination of mesalazine in clinically and environmentally relevant samples.

4. Conclusion

This work presents an ultrasensitive and highly selective voltammetric method for the determination of the anti-inflammatory drug mesalazine using the Co-MOR/MC-GCE sensor. The application of a nanocomposite based on Co-exchanged zeolite mordenite and mesoporous carbon in the simple and low-cost drop-casting procedure of the GCE surface modification resulted in the fabrication of a unique MES sensing platform characterized by the very low limit of detection, a high selectivity, short- and long-term stability, as well as excellent repeatability and reproducibility.

The performed CV measurements demonstrate that the electrode reaction on the Co-MOR/MC-GCE is realized via diffusion-controlled, two-electron and two-proton, reversible oxidation step. Through optimization of the instrumental parameters of the DVP technique and the pH value of the supporting electrolyte, the presented protocol distinguishes itself with the low limit of detection equal to $1.76 \cdot 10^{-9}$ mol L^{-1}. The nanomolar level of *LOD*, achieved without any preconcentration step, creates the possibility of determining the trace amount of mesalazine in samples with different kinds of matrices. No influences from various inorganic ions, basic excipients of drugs, surfactants, and acids (uric, humic) ensures, at the same time, the high selectivity of MES determination without special sample preparation. These findings were confirmed by the analysis of MES as a main ingredient of pharmaceuticals, as well as its trace amount in environmental waters and biological fluids, such as human urine and serum.

CRediT authorship contribution statement

Katarzyna Fendrych: Conceptualization, Methodology, Investigation, Validation, Visualization, Formal analysis, Writing – original draft, Writing – review & editing. **Radosław Porada:** Visualization, Formal analysis, Writing – original draft, Writing – review & editing. **Bogusław Baś:** Conceptualization, Supervision, Resources, Writing – review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

BB reports financial support was provided by AGH University of Science and Technology. KF reports financial support was provided by Ministry of Education and Science for the AGH University of Science and Technology in Kraków (Project No 16.16.160.557).

Data availability

Data will be made available on request.

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